

1. A 25-year-old woman comes to the physician at her husband's request because of increasing irritability and tense mood over the past year. Her husband would like to start a family, and he is concerned about her ability to handle the added responsibility. She says that she worries about their finances despite the fact that they have no debt and she and her husband have jobs that pay well. Her pulse is 84/min, and blood pressure is 120/70 mm Hg. Physical examination shows no abnormalities except for sweaty palms. Mental status examination shows tension and anxiety. She admits to occasional difficulty sleeping and periods of mild depression but does not think that this is out of the ordinary. Laboratory studies show:

Hematocrit	38%
Erythrocyte count	4 million/mm ³
Leukocyte count	5500/mm ³
Serum	
Thyroid-stimulating hormone	0.5 μU/mL
Thyroxine (T ₄)	12 μg/dL

Toxicology screening is negative. Which of the following is the most appropriate pharmacotherapy for this patient?

- ☐ A) Bupropion
- ☐ B) Buspirone
- ☐ C) Levothyroxine
- ☐ D) Methimazole
- ☐ E) Quetiapine



Next



Lab Values



Review



Help



Pause

2. A 13-year-old girl is brought to the physician by her father for a well-child examination. He is concerned about her weight and eating habits. He says that she is always "on the go" and never sits down for a full meal. She will often just grab a piece of fruit when she is on her way to an activity. She will eat a full meal when the family has dinner together every Sunday. She has talked about wanting to become a vegetarian because of her concern for animals. She has had the same group of friends since elementary school. She is the captain of her soccer team and practices 4 days weekly. She is active in an after-school drama program 3 days weekly and is the lead in this season's play. She maintains a B grade average. Her father says that she talks on the telephone "constantly" and is animated and cheerful most of the time. When she is in her bedroom, she prefers to keep the door closed and stops talking if she is on the telephone and her father walks into the room. She has a disorganized bedroom and often falls fast asleep on a bed covered with piles of clothes. Menarche has not occurred. She appears thin and muscular. She is at the 50th percentile for height and 35th percentile for weight, which is unchanged from last year. Vital signs are within normal limits. Breast and pubic hair development are Tanner stage 2. Physical examination shows no abnormalities. When interviewed alone, she is animated. She shrugs and rolls her eyes when asked about her diet and weight. She thinks she is fine and does not understand why her father is so upset. Which of the following is the most appropriate next step?
- ☐ A) Reassure the father that this is normal development
 - ☐ B) Recommend individual psychotherapy
 - ☐ C) Recommend nutritional counseling
 - ☐ D) Recommend that the father keep a log of what his daughter eats
 - ☐ E) Schedule weekly examination and weighing

3. A 32-year-old woman comes to the physician for a follow-up examination 6 months after she was a passenger on an airplane that skidded off the runway into the harbor. Two people were killed; she escaped without serious injury. After the crash, she had insomnia, a depressed mood, nightmares about plane crashes, and daytime flashbacks that have been treated with clonazepam and paroxetine. Overall, her symptoms have improved, but she is still too anxious to fly. She works as a buyer for a large department store and was told by her manager that her job is in jeopardy because she is required to travel to Europe monthly. Physical examination shows no abnormalities. Mental status examination shows mild anxiety. Her thoughts are organized and goal directed. She has difficulty completing serial seven subtractions. In addition to continuing clonazepam and paroxetine therapy, which of the following is the most appropriate next step in management?

- ☐ A) Recommend that the patient switch to a job that does not require air travel
- ☐ B) Begin cognitive behavioral therapy
- ☐ C) Begin group therapy
- ☐ D) Begin psychodynamic psychotherapy
- ☐ E) Add quetiapine to the current regimen

4. A 27-year-old teacher comes to the physician requesting a prescription for sleep medication. She has been unable to fall asleep until 2 or 3 am since the start of the school year 3 months ago. When she awakens at 6 am, she feels drowsy and remains tired throughout the morning. Last week, she fell asleep at her desk while her students were watching a movie. She believes that her job is in jeopardy. Drinking one glass of wine before going to bed has not helped her fall asleep. She recently took one of her mother's zolpidem tablets; she only slept for 2 hours and was then awake until 3 or 4 am. During vacations and on weekends, she typically sleeps from 3 am until noon and feels energetic and well rested. She currently does not take any medications or drink alcohol. Physical examination shows no abnormalities. On mental status examination, she appears tired and is teary. She can recall two of three objects after 5 minutes. Which of the following is the most likely diagnosis?

- ☐ A) Adjustment disorder with depressed mood
- ☐ B) Circadian rhythm sleep disorder
- ☐ C) Dysthymic disorder
- ☐ D) Major depressive disorder
- ☐ E) Narcolepsy
- ☐ F) Primary insomnia
- ☐ G) REM sleep behavior disorder

5. A 42-year-old woman, gravida 1, para 1, comes to the physician because of a 5-month history of increasingly severe episodes of lower abdominal pain. The pain is exacerbated during bowel movements, sexual intercourse, and menses. Treatment with several analgesics has not controlled the pain. Laparoscopy 4 years ago and a second laparoscopy 2 years ago for evaluation of pain showed no abnormalities. She is sexually active with one male partner, and they use condoms for contraception. She is a single parent and lost her job 6 months ago. She now has difficulty sleeping 1 to 3 nights each week because she worries about finding a job. Abdominal examination shows tenderness in both lower quadrants. Pelvic examination shows a mobile, nontender uterus and normal, nontender adnexa. A complete blood count, erythrocyte sedimentation rate, Pap smear, and urine and cervical cultures show no abnormalities. Which of the following is the most appropriate next step to determine the cause of this patient's pain?

- ☐ A) MRI of the pelvis
- ☐ B) Psychiatric assessment
- ☐ C) Hysteroscopy
- ☐ D) Laparoscopy
- ☐ E) Endometrial biopsy

6. A 57-year-old woman comes to the physician because of difficulty sleeping, tearfulness, and restlessness since her daughter was diagnosed with metastatic breast cancer 3 days ago. She reports that when she goes to bed at night, she is unable to fall asleep for several hours and lays in bed worrying about her daughter's situation. The patient underwent a mastectomy for breast cancer 7 years ago. She takes acetaminophen/butalbital for occasional migraines. Her vital signs are within normal limits. Physical examination shows no abnormalities. On mental status examination, she is tearful and tense but calms during the conversation. There is no evidence of suicidal ideation. Which of the following is the most appropriate next step in management?

- ☐ A) Biofeedback
- ☐ B) Carbamazepine therapy
- ☐ C) Clonazepam therapy
- ☐ D) Clonidine therapy
- ☐ E) Imipramine therapy
- ☐ F) Olanzapine therapy
- ☐ G) Pentobarbital therapy
- ☐ H) Sertraline therapy

7. A 25-year-old woman is brought to the physician by the owner of the stable where she works because she has been talking to the horses for 7 days. She dropped out of high school at the age of 17 years and has worked at the stable since that time. She lives alone and has no friends. On questioning, she states that the horses tell her how to lead her life and that they have been given special powers by a god. Which of the following is the most likely diagnosis?

- ☐ A) Autistic disorder
- ☐ B) Avoidant personality disorder
- ☐ C) Delusional disorder
- ☐ D) Schizoid personality disorder
- ☐ E) Schizophrenia

8. A 7-year-old boy is brought to the physician by his adoptive parents because of ongoing concern about his behavior and academic performance. He is two grade levels behind his peers in reading and frequently has fights with other children. He has no history of serious illness and takes no medications. His biological mother abused multiple substances before and during pregnancy. On adoption at the age of 15 months, there was some evidence that she may have physically abused him. He is at the 10th percentile for height and 5th percentile for weight. His pulse is 82/min, and blood pressure is 116/78 mm Hg. Examination shows a flattened nasal bridge, a long philtrum, and prominent ears. During the interview, he is friendly but becomes frustrated and distracted and is unable to sit still. Which of the following is the most likely diagnosis?

- ☐ A) Angelman syndrome
- ☐ B) Fetal alcohol syndrome
- ☐ C) Fragile X syndrome
- ☐ D) Rett disorder
- ☐ E) Shaken baby syndrome

9. A 17-year-old boy is brought to the emergency department by his mother because of a 20-minute episode of severe chest pain and sweating that started 1 hour ago while he was watching television. On arrival, he is alert and repeatedly expresses concern that something is "very wrong." He reports that during the episode, his heart was "pounding," he felt short of breath, and he was sure he was having a heart attack and was going to die. One month ago, he had a similar episode that occurred in the evening approximately 30 minutes after his high school football practice. At that time, results of laboratory studies and an ECG obtained in the emergency department showed no abnormalities. After the initial episode, he quit the football team, and he has continued to limit his physical activity because he is afraid of bringing on another episode. He reports no insomnia or changes in appetite. His mother states that he was diagnosed with an innocent heart murmur at the age of 6 months, but it has not been noted on examinations since the age of 1 year. His paternal grandfather died of a myocardial infarction 3 years ago at the age of 68 years. The patient takes no medications. He does not smoke cigarettes, drink alcohol, or use illicit drugs. He performs well in school. He is 180 cm (5 ft 11 in) tall and weighs 73 kg (161 lb); BMI is 23 kg/m². His temperature is 37°C (98.6°F), pulse is 98/min, respirations are 20/min, and blood pressure is 130/83 mm Hg. Cardiac examination shows a normal S₁ and S₂ with variation of S₂ on respiration. The remainder of the examination shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Acute stress disorder
- ☐ B) Delusional disorder
- ☐ C) Generalized anxiety disorder
- ☐ D) Hypertrophic cardiomyopathy
- ☐ E) Idiopathic recurrent pericarditis
- ☐ F) Panic disorder
- ☐ G) Paroxysmal supraventricular tachycardia

10. A previously healthy 27-year-old woman comes to the physician because of a 2-month history of crying spells, poor concentration, fatigue, and lethargy. She says that she sleeps at least 12 hours daily and is no longer interested in socializing. She recently broke up with her boyfriend of 10 months, and her work performance has begun to decline; she is afraid that she will lose her job if it does not improve. She does not drink alcohol or use illicit drugs. Menses occur at regular 28- to 30-day intervals. Her last menstrual period was 2 weeks ago. She has had a 9-kg (20-lb) weight gain over the past 9 months. She is 168 cm (5 ft 6 in) tall and weighs 66 kg (145 lb); BMI is 23 kg/m². Physical examination shows no abnormalities. Mental status examination shows a sad mood and restricted affect. Which of the following is the most likely diagnosis?

- ☐ A) Adjustment disorder with depressed mood
- ☐ B) Dysthymic disorder
- ☐ C) Generalized anxiety disorder
- ☐ D) Major depressive disorder
- ☐ E) Premenstrual dysphoric disorder

11. A 25-year-old primigravid woman at 16 weeks' gestation has had increased episodes of crying, difficulty falling asleep, decreased energy, and loss of appetite over the past 6 weeks. She has a history of recurrent major depressive disorder. Her family is concerned that she might inadvertently harm herself or her fetus. Her husband reports that she has lost interest in daily activities. She states that she has no suicidal thoughts. She is concerned about her ability to take care of a newborn and is questioning her decision not to terminate the pregnancy. Which of the following is the most appropriate next step in management?

- ☐ A) Reassurance with return visit in 2 weeks
- ☐ B) Explanation of the risks and benefits of antidepressant therapy
- ☐ C) Explanation of the risks and benefits of therapeutic abortion
- ☐ D) Support group therapy
- ☐ E) Admission to the hospital

The response options for the next **2** items are the same. Select one answer for each item in the set.

For each patient with depressed mood, select the most likely diagnosis.

- ☐ A) Adjustment disorder with depressed mood
- ☐ B) Bereavement
- ☐ C) Bipolar disorder, depressed
- ☐ D) Cyclothymic disorder
- ☐ E) Dysthymic disorder
- ☐ F) Major depressive disorder
- ☐ G) Mood disorder due to a general medical condition
- ☐ H) Substance-induced mood disorder

12. A previously healthy 37-year-old man comes to the physician because of a 6-month history of depressed mood, fatigue, poor concentration, and difficulty sleeping. He works as an accountant and has had difficulty staying awake at work, particularly in the afternoon. He says that his wife sleeps in a separate room because of his loud snoring. He drinks alcohol at social functions only. He is 175 cm (5 ft 9 in) tall and weighs 109 kg (240 lb); BMI is 35 kg/m². Physical examination shows no other abnormalities. Mental status examination shows a mildly depressed mood.

For each patient with depressed mood, select the most likely diagnosis.

- ☐ A) Adjustment disorder with depressed mood
- ☐ B) Bereavement
- ☐ C) Bipolar disorder, depressed
- ☐ D) Cyclothymic disorder
- ☐ E) Dysthymic disorder
- ☐ F) Major depressive disorder
- ☐ G) Mood disorder due to a general medical condition
- ☐ H) Substance-induced mood disorder

13. A 23-year-old graduate student comes to the physician because of depressed mood and feelings of hopelessness since breaking up with her boyfriend 1 week ago. During this period, she has rarely left her apartment, has only eaten a small amount of food, and has isolated herself from her friends and family. She has been drinking more alcohol than normal during this period. Physical examination shows no abnormalities. On mental status examination, she becomes tearful when she describes the breakup. When asked about her mood, she says, "I'm miserable. He was my life!" There is no evidence of suicidal ideation.

14. A 37-year-old male police officer comes to the physician at the request of his superiors 1 week after he witnessed a terrorist bombing during which several civilians and three fellow police officers were killed. He sustained only minor injuries and assisted in rescuing survivors and gathering body parts. Since the bombing, he has felt emotionally numb and has been unable to enjoy activities he used to find pleasurable. He has continued to work but has requested assignments far removed from the site of the attack. He describes his sleep as fitful. Two years ago, he sustained a bullet wound to the right calf, which left him with a slight limp. He has no other history of medical or psychiatric illness. He says he used to drink one to two beers nightly but has been drinking two to three beers nightly for the past week. Physical examination confirms the previous gunshot injury to the calf and shows no other abnormalities. On mental status examination, he is irritable and says he is not depressed. He tells the physician, "They made me come. I'm not interested in talking with anyone. I just want to spend time with my buddies and be left alone." Which of the following is the most appropriate initial step in management?

- ☐ A) Encourage the patient to discuss the trauma in detail
- ☐ B) Provide information about the range of reactions to trauma
- ☐ C) Recommend group therapy with other trauma survivors
- ☐ D) Recommend a 1-month medical leave of absence from active duty
- ☐ E) Recommend a physician-led trauma debriefing series
- ☐ F) Recommend a 12-step program
- ☐ G) Begin clonazepam therapy
- ☐ H) Begin fluoxetine therapy

15. A 21-year-old man comes to the physician because of a 3-week history of muscle aches and fatigue associated with exercise. He is worried that he might have multiple sclerosis. Two months ago, he was concerned that a mole on his right arm may be cancerous. The physician reassured him that the mole was nonmalignant. One month ago, he had lateral chest pain while jogging. An ECG at that time showed no abnormalities. He appeared relieved once he heard the results. Today, his temperature is 36.7°C (98°F), pulse is 70/min, respirations are 20/min, and blood pressure is 130/84 mm Hg. Physical examination shows no abnormalities. On mental status examination, he appears tense but makes full eye contact and is cooperative. He speaks clearly and at a normal rate, and his thought process is goal directed. Which of the following is the most likely diagnosis?

- ☐ A) Conversion disorder
- ☐ B) Delusional disorder, somatic type
- ☐ C) Factitious disorder
- ☐ D) Hypochondriasis
- ☐ E) Somatization disorder

16. A 2½-year-old boy is brought by his parents for a well-child examination. He recently started attending day care. The father reports that his son protests and hangs on his coat sleeves each morning so the day-care provider cannot take him. He sometimes says he has a stomachache. The day-care provider reports that after the father drives away, the child settles down and plays happily with other children. He eats well and takes a 90-minute nap. When his mother comes to pick him up, he whines, stamps his feet, clings to the day-care provider, and demands cookies. Physical examination shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Adjustment disorder with anxiety
- ☐ B) Adjustment disorder with depressed mood
- ☐ C) Dysthymic disorder
- ☐ D) Factitious disorder
- ☐ E) Generalized anxiety disorder
- ☐ F) Major depressive disorder
- ☐ G) Oppositional defiant disorder
- ☐ H) Panic disorder
- ☐ I) Parent-child relational problem
- ☐ J) Post-traumatic stress disorder
- ☐ K) Separation anxiety disorder
- ☐ L) Sleep terror disorder
- ☐ M) Normal reaction

17. A 52-year-old homeless man with alcoholism is brought to the emergency department by police after being found agitated and confused. He says he feels nervous and could use a drink. His temperature is 37°C (98.6°F), pulse is 120/min, respirations are 20/min, and blood pressure is 138/86 mm Hg. He is trembling. His hematocrit is 30%. Urinalysis shows 7 WBC/hpf and many oxalate crystals. Which of the following is the most likely cause of this patient's symptoms?

- ☐ A) Ethylene glycol toxicity
- ☐ B) Lactic acidosis
- ☐ C) Methanol toxicity
- ☐ D) Paraldehyde toxicity
- ☐ E) Sepsis

18. A 5-year-old girl is brought to the physician for a well-child examination. Her mother is concerned because she uses nonsensical and repetitive words. Her teacher reports that other children do not like to play with her, and they laugh at her because she often flaps her hands when she speaks. She frequently spends time alone in her room twirling on her toes. Her mother drank two beers daily and used cocaine occasionally during the pregnancy. The patient is at the 60th percentile for height and 55th percentile for weight. Her pulse is 76/min, and blood pressure is 100/80 mm Hg. Physical examination shows eczema and bruising over the knees. Mental status examination shows a neutral mood. She avoids eye contact and refers to the physician as a "minzer." Which of the following is the most likely explanation for these findings?

- ☐ A) Attention-deficit/hyperactivity disorder
- ☐ B) Fetal alcohol syndrome
- ☐ C) Pervasive developmental disorder
- ☐ D) Phenylketonuria
- ☐ E) Normal development

19. A 47-year-old woman is brought to the emergency department by her husband because of increasing confusion during the past 2 days. On arrival, she has a generalized tonic-clonic seizure lasting 4 minutes. She has bipolar disorder treated with several medications, but her husband is unsure of their names. He knows that she sometimes takes ibuprofen for mild arthritis pain caused by exercise. He says that she has been active and exercising more lately, but he cannot think of other changes in her routine. She has no history of seizure disorder. She is oriented to person but not to place and time. Her pulse is 90/min, and blood pressure is 140/90 mm Hg. On physical examination, she is tremulous and somnolent. There is bilateral nystagmus. An ECG shows a second-degree atrioventricular block. The most likely cause of this patient's symptoms is an adverse effect of which of the following medications?

- ☐ A) Bupropion
- ☐ B) Haloperidol
- ☐ C) Lithium
- ☐ D) Risperidone
- ☐ E) Valproic acid

20. A 28-year-old man is brought to the emergency department because of flu-like symptoms and cough productive of clear sputum for 24 hours. He has been hospitalized eight times over the past 5 years because of bizarre delusions and hallucinations during which a voice tells him what to do. He has a 9-year history of type 1 diabetes mellitus. In addition to insulin, he is receiving clozapine therapy (400 mg daily) for his psychotic condition. During mental status examination, he is hallucinating and says that he is being pursued by special agents of the government. Leukocyte count is $2750/\text{mm}^3$, and serum glucose concentration is 140 mg/dL. Urinalysis is within normal limits. Which of the following is the most appropriate next step in management?

- ☐ A) Add lithium carbonate therapy
- ☐ B) Add penicillin therapy
- ☐ C) Decrease the dose of clozapine
- ☐ D) Discontinue clozapine therapy
- ☐ E) Increase the dose of clozapine

21. A 7-year-old boy is brought to the physician by his mother because of difficulty at school since beginning first grade 6 weeks ago. His teachers state that he has been daydreaming frequently, does not complete his work, and does not participate in classroom discussions. His mother reports that he is very shy, will not go to sleepovers, and has stomachaches every weekday morning. He played on a soccer team, but he quit when his mother could not drive him to games and he had to carpool with another family. Physical examination shows no abnormalities. On mental status examination, the boy sits on his mother's lap, does not make eye contact with the physician, and is hesitant to answer questions but seems eager to please. Which of the following is the most likely explanation for these findings?

- ☐ A) Attention-deficit/hyperactivity disorder
- ☐ B) Dysthymic disorder
- ☐ C) Generalized anxiety disorder
- ☐ D) Separation anxiety disorder
- ☐ E) Social phobia
- ☐ F) Age-appropriate behavior

22. A 27-year-old man is brought to the emergency department 30 minutes after his brother found him agitated, tremulous, and complaining of a headache. He had eaten a peanut butter sandwich 12 hours before his symptoms began. He has a 10-year history of severe major depressive disorder. During this period, he has attempted suicide three times despite treatment with both selective serotonin reuptake inhibitors and tricyclic antidepressants. Eight days ago, he went to a new physician because of increasing despondency. This physician immediately discontinued the fluoxetine and substituted tranylcypromine to start 1 week later. He also instructed the patient to start a monoamine oxidase inhibitor diet immediately. Current medications also include diphenhydramine as needed for seasonal allergies. He has no history of hypertension, diabetes mellitus, or migraine. On arrival, he is agitated. His temperature is 40.6°C (105°F), pulse is 130/min, respirations are 26/min, and blood pressure is 180/120 mm Hg. Physical examination shows diaphoresis and tremor of the upper extremities. Deep tendon reflexes are 3+ bilaterally; there are no pathologic reflexes. On mental status examination, he is complaining of occipital headache. He is oriented to person but not to place or time. Which of the following is the most likely cause of these findings?

- ☐ A) Acute anxiety attack
- ☐ B) Acute fluoxetine withdrawal
- ☐ C) Drug-drug interaction
- ☐ D) Ingestion of peanut butter
- ☐ E) Use of diphenhydramine

23. An 82-year-old man with dementia, Alzheimer type, is brought to the physician because of a 4-day history of visual hallucinations and irritability. He has a 2-month history of insomnia. He has osteoarthritis, hypertension, and hypothyroidism. His medications are rivastigmine, ibuprofen, atenolol, levothyroxine, and amitriptyline. His blood pressure is 128/88 mm Hg. Physical examination shows no abnormalities. On mental status examination, he recalls zero of three objects after 5 minutes. During the examination, he points his finger and sternly says, "You need to stop doing that right this minute." When asked whom he is addressing, he says, "that tiny little man shouting over in the corner." Results of a complete blood count, measurement of serum glucose concentration, and kidney and liver function tests are within the reference ranges. Urinalysis shows no abnormalities. Discontinuation of which of the following of this patient's medications is most appropriate at this time?

- ☐ A) Amitriptyline
- ☐ B) Atenolol
- ☐ C) Ibuprofen
- ☐ D) Levothyroxine
- ☐ E) Rivastigmine

24. In a fit of anger, a 22-year-old woman slaps her 2-year-old child across the face. There is no other history of abuse. Which of the following is the most likely outcome of this experience for the child?

- ☐ A) Brief psychotic disorder
- ☐ B) Dysthymic disorder
- ☐ C) Pain disorder
- ☐ D) Post-traumatic stress disorder
- ☐ E) No impact on development

25. A 20-year-old woman comes to student health services because of a 6-week history of difficulty sleeping. She says that two to three times weekly, she wakes up during the night in the middle of a vivid dream that usually involves the need to escape from a physical threat such as a rapist. When she tries to run in her dreams, her legs feel progressively heavier until she is barely able to move; it is at this point that she wakes up. She says that she has difficulty falling back to sleep; during the past 2 weeks, she has been drinking a glass of wine to aid her in falling back to sleep. She feels tired throughout the day and is concerned that her grades will begin to decline. Three months ago, she broke up with her boyfriend of 1 year because he refused to stop using marijuana. She has not dated since the breakup, but she spends time with friends. Physical examination shows no abnormalities. On mental status examination, she is friendly and cooperative. She appears mildly anxious, but she smiles and laughs on occasion. She says she is tired. Her thought process is logical and goal directed. She is oriented to person, place, and time. There is no evidence of auditory or visual hallucinations. Which of the following is the most likely diagnosis?

- ☐ A) Adjustment disorder with anxiety
- ☐ B) Dysthymic disorder
- ☐ C) Generalized anxiety disorder
- ☐ D) Nightmare disorder
- ☐ E) Primary insomnia
- ☐ F) Sleep terror disorder

26. An 82-year-old man is brought to the physician by his daughter because he has been forgetful during the past 6 weeks. She is particularly concerned because a small fire started recently when he left a pot unattended on the stove. The patient's wife died 1 year ago, and he says he has been distracted recently and cannot stop thinking about her. He lives alone and pays his bills, does routine household chores, and drives short distances for errands. His daughter says he was doing well on his own until recently. He is 168 cm (5 ft 6 in) tall and weighs 63 kg (140 lb); BMI is 23 kg/m². His temperature is 37°C (98.6°F), pulse is 78/min and regular, respirations are 18/min, and blood pressure is 130/80 mm Hg. Physical examination shows no abnormalities. On mental status examination, he is alert and cooperative. He becomes very sad when he talks about his wife's death. Which of the following is the most appropriate next step in management?

- ☐ A) Follow-up examination in 3 months
- ☐ B) Mini-Mental State Examination
- ☐ C) Recommend assisted living
- ☐ D) CT scan of the head
- ☐ E) Begin donepezil therapy

27. A 6-year-old boy is brought to the physician because of difficulty in school since he began first grade 2 months ago. He tells his parents that he hates school and does not want to go. He did not attend preschool. His first-grade teacher describes him as a daydreamer who does not pay attention and will not read or write. She reports that he has no friends and does not listen to her. She admits that she is getting quite frustrated with him because he will not look at her when she speaks to him. She states that he is good with numbers and prefers math to reading and spelling. She has noticed that he fills his notebook with drawings of haystacks that look identical. She says that he does not seem to be happy in school. Recent psychoeducational testing showed a performance IQ of 90 and a verbal IQ of 90. Physical examination shows no abnormalities. Mental status examination shows no evidence of psychosis, mania, or depression. Which of the following is the most likely diagnosis?

- ☐ A) Asperger disorder
- ☐ B) Expressive language disorder
- ☐ C) Mild mental retardation
- ☐ D) Reading disorder
- ☐ E) Separation anxiety disorder
- ☐ F) Normal development

28. A 62-year-old woman comes to the physician because of loss of libido for 9 months. She was sexually active with her husband until he died 3 years ago. Since meeting her 58-year-old male friend 3 months ago, she has had consistent excuses for not engaging in sexual intercourse. She feels guilty about starting a new relationship so soon after her husband's death and has had recurrent dreams over the past 6 months about her husband and their happy marriage. She has not had crying spells or difficulty with sleep or appetite. She has been receiving hormone replacement therapy since undergoing a total abdominal hysterectomy and bilateral salpingo-oophorectomy 5 years ago. Physical examination shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Hypoactive sexual desire disorder
- ☐ B) Sexual arousal disorder
- ☐ C) Substance-induced sexual dysfunction
- ☐ D) Vaginismus
- ☐ E) Normal aging

29. An 18-year-old man is brought to the emergency department by friends 1 hour after they found him on the couch at a party unable to move. Earlier that evening, he had been using synthetic heroin. On arrival, his pulse is 85/min, respirations are 18/min, and blood pressure is 130/80 mm Hg. Examination shows stiffness in all extremities, drooling, and slow response to questions. Which of the following brain regions is most likely affected?

- ☐ A) Cerebellar vermis
- ☐ B) Globus pallidus
- ☐ C) Locus ceruleus
- ☐ D) Mammillary bodies
- ☐ E) Nucleus accumbens
- ☐ F) Substantia nigra

30. A 32-year-old woman comes to the physician because of increasingly severe pain that originates in her left shoulder and radiates to her elbow. She describes the pain as constant and burning, rating her current pain as a 7 on a 10-point scale. Eighteen months ago, she sustained a nerve injury of the left upper extremity in a motor vehicle collision. Since that time, she has been unable to return to work. Current medications include oxycodone and gabapentin. Physical examination shows atrophy of the left thenar eminence. Muscle strength in the left forearm and finger flexors is 3/5. On sensory examination, there is severe pain with light stroking of the anterolateral aspect of the left arm. Further sensory testing is deferred. During the examination she tells her physician, "I'm tired of all this. My medication is not strong enough. It only takes the edge off my pain, which is only getting worse. I'm realizing I'll be like this forever." Which of the following is the most appropriate response by the physician?

- ☐ A) "Are you worried about more nerve damage developing?"
- ☐ B) "Do you ever use more pain medication than is prescribed?"
- ☐ C) "Have you been feeling like just giving up?"
- ☐ D) "Is the pain caused by touch socially limiting?"
- ☐ E) "Let's review your medical concerns."

31. An 8-year-old girl is brought to the physician by her mother because of frequent stomachaches over the past 3 months. She has a 1-year history of difficulty falling asleep and poor concentration. She has always been a good student and excels in extracurricular activities. Her parents divorced 6 months ago. Physical examination shows no abnormalities. Mental status examination shows an anxious affect and coherent thought process. She says that she is worried about her classes and about her performance in the school band and on the soccer team. Which of the following is the most likely diagnosis?

- ☐ A) Generalized anxiety disorder
- ☐ B) Obsessive-compulsive disorder
- ☐ C) Panic disorder
- ☐ D) Post-traumatic stress disorder
- ☐ E) Separation anxiety disorder

The response options for the next **2** items are the same. Select one answer for each item in the set.

For each patient with anxiety, select the most appropriate pharmacotherapy.

- ☐ A) Bupropion
- ☐ B) Buspirone
- ☐ C) Chlorpromazine
- ☐ D) Fluoxetine
- ☐ E) Haloperidol
- ☐ F) Imipramine
- ☐ G) Lorazepam
- ☐ H) Perphenazine
- ☐ I) Propranolol

32. A 27-year-old man comes to the physician because of anxiety about a major speech that he must deliver in 3 days. He is excited about the opportunity for his career but is concerned that his apprehension and great fear of public speaking will affect his performance. He requests a tranquilizer to help with his anxiety. He has no history of medical problems. He has a family history of alcohol dependence. He does not drink alcohol or use drugs. His pulse is 76/min, respirations are 12/min, and blood pressure is 130/82 mm Hg. Examination shows normal findings.

For each patient with anxiety, select the most appropriate pharmacotherapy.

- ☐ A) Bupropion
- ☐ B) Buspirone
- ☐ C) Chlorpromazine
- ☐ D) Fluoxetine
- ☐ E) Haloperidol
- ☐ F) Imipramine
- ☐ G) Lorazepam
- ☐ H) Perphenazine
- ☐ I) Propranolol

33. A 27-year-old man comes to the physician because of anxiety about a major speech that he must deliver in 3 days. He has a great fear of public speaking and is convinced that his apprehension and tremulous delivery will damage his performance. He requests a tranquilizer to help with his anxiety. He takes theophylline and uses corticosteroid and albuterol inhalers for asthma. He has a family history of alcohol dependence. He does not drink alcohol or use drugs. His pulse is 66/min, respirations are 12/min, and blood pressure is 132/80 mm Hg. Examination shows normal findings.

34. A 47-year-old man is brought to the physician by his wife because of "unbearable" behavior during the past 2 weeks. His wife reports that he has been talking about his various inventions constantly and recently bought a new recreational vehicle, even though they do not travel long distances. He repeatedly tells his wife that "the time to make and enjoy money has come." He has been openly critical of their pastor's prayers during church services. He claims that the pastor does not know what he is talking about because "God is talking to me, not him." His wife says that he has had episodes of similar behavior during the past 10 years that have been more tolerable. He has peptic ulcer disease treated with ranitidine and hypertension treated with triamterene. He does not smoke or drink alcohol. He is 170 cm (5 ft 7 in) tall and weighs 82 kg (180 lb); BMI is 28 kg/m². His pulse is 80/min, and blood pressure is 150/95 mm Hg. Physical examination shows no other abnormalities. On mental status examination, he is relaxed and talkative, jumping from one topic to another. He claims that his wife does not understand him and that she has been plotting with the pastor against him. He denies any problems and says that he feels great and is capable of great things. Laboratory findings are within the reference range. Urine toxicology screening is positive for marijuana. Which of the following is the most likely diagnosis?

- ☐ A) Bipolar disorder
- ☐ B) Delusional disorder
- ☐ C) Mood disorder due to a general medical condition
- ☐ D) Schizophrenia
- ☐ E) Substance-induced mood disorder

35. A 32-year-old woman is hospitalized for evaluation of a 5-month history of headaches. While recovering from sedation with lorazepam for cerebral angiography, she has dull, aching abdominal pain that is localized to the right upper quadrant of the abdomen without radiation. It has occurred several times over the past week after meals. She has had no nausea or vomiting. Following a CT scan 3 months ago, she had aphonia for 48 hours. She has a history of cholecystectomy, appendectomy, hysterectomy, and exploratory laparoscopy for endometriosis. Cerebral angiography shows normal findings. Repeat neurologic examinations and CT scan of the head are negative. Her temperature is 36.4°C (97.6°F), pulse is 68/min, respirations are 12/min, and blood pressure is 122/76 mm Hg. During examination, she complains of unbearable pain, although the abdomen is tender to light palpation without guarding or rebound. No masses are palpated. Bowel sounds are normal. Pelvic examination shows normal findings. Test of the stool for occult blood is negative. Which of the following is the most likely diagnosis?

- ☐ A) Acute intermittent porphyria
- ☐ B) Acute pancreatitis
- ☐ C) Conversion disorder
- ☐ D) Small-bowel obstruction
- ☐ E) Somatization disorder

36. A 42-year-old woman comes to the physician because of a 6-month history of intrusive thoughts of her family being harmed. She says that the only way to stop the thoughts is to wash her hands 10 times and count backwards from 100. The thoughts occur 15 to 20 times throughout the day. She takes no medications. Physical examination shows no abnormalities except for dry, chapped hands. Mental status examination shows a mildly anxious affect. There is no evidence of hallucinations or delusions. She is alert and oriented to person, place, and time. Concentration and short- and long-term memory are intact. Which of the following is the most appropriate next step in management?

- ☐ A) Psychodynamic psychotherapy only
- ☐ B) Diazepam and behavioral therapy
- ☐ C) Diazepam and psychodynamic psychotherapy
- ☐ D) Diazepam therapy only
- ☐ E) Fluvoxamine and behavioral therapy
- ☐ F) Fluvoxamine and psychodynamic psychotherapy
- ☐ G) Fluvoxamine therapy only

37. A 24-year-old man is brought to the emergency department from a psychiatric hospital because of a decreasing level of consciousness over the past 3 days. He has become progressively less responsive and has remained in bed for the past 12 hours without speaking or following commands. He has a history of schizophrenia with multiple psychiatric hospitalizations and is currently being treated with haloperidol and benztropine. His temperature is 39.4°C (103°F), pulse is 120/min, respirations are 28/min, and blood pressure is 180/100 mm Hg. He withdraws weakly to noxious stimuli. His eyes are open, but he does not fix or follow visual stimuli or gaze around the room. He is mute and does not follow commands. Examination shows marked rigidity of the neck and all extremities. Laboratory studies show:

Leukocyte count	18,000/mm ³
Segmented neutrophils	80%
Lymphocytes	20%
Serum creatine kinase	12,000 U/L
Cerebrospinal fluid	
Erythrocyte count	2/mm ³
Leukocyte count	1/mm ³
Protein	28 mg/dL
Glucose	78 mg/dL

A CT scan of the head shows no abnormalities. Which of the following is the most likely underlying cause?

- ☐ A) Catatonia
- ☐ B) Conversion reaction
- ☐ C) Dissociative fugue
- ☐ D) Drug reaction
- ☐ E) Infection
- ☐ F) Malingering
- ☐ G) Toxin

38. Five weeks after being discharged from the hospital after treatment for a psychotic episode, a 27-year-old man comes to the physician for a follow-up examination. During hospitalization, he claimed that he was instructed by the Lord to found a new religion and that a famous gospel singer was in love with him. Today, he says that he still hears the voices of the Lord and members of the church he attends in his apartment and when he shops in the supermarket. He no longer believes the world needs a new religion because the Lord is no longer instructing him to create one. He states, "My father in heaven tells me that he is at peace, and therefore, so am I." Current medications include risperidone and lorazepam. He drinks one to two beers on weekends. He used marijuana regularly in college but has abstained for the past 5 years. He appears clean and is casually dressed. His temperature is 36.7°C (98°F), pulse is 72/min, respirations are 20/min, and blood pressure is 130/72 mm Hg. Physical examination shows no abnormalities. Mental status examination shows a calm affect. He is cooperative, alert, and oriented to person, place, and time. Based on this information, which of the following is the most likely current diagnosis for this patient?

- ☐ A) Bipolar disorder
- ☐ B) Cyclothymic disorder
- ☐ C) Delusional disorder
- ☐ D) Schizoaffective disorder
- ☐ E) Substance-induced mood disorder

39. A 32-year-old financial analyst comes to the physician at her employer's request because of bizarre behavior and poor work productivity over the past 6 months. She has been arriving late, leaving early, and frequently taking long breaks. She has had daily mood swings; sometimes she appears withdrawn, lethargic, and belligerent and other times she appears energetic and loquacious. Over the past 2 weeks, she has had several arguments with coworkers in which she accused them of stealing or sabotaging her work. This morning, she threatened to physically assault her supervisor after he questioned her about a 2-hour absence from work. Her temperature is 37°C (98.6°F), pulse is 90/min, respirations are 16/min, and blood pressure is 160/100 mm Hg. The pupils are dilated and reactive to light. Deep tendon reflexes are increased. On mental status examination, she is angry but cooperative. She describes her supervisor's unfairness and the unfriendliness of other analysts; she suspects that they are jealous of her. She speaks in a loud, rapid voice, frequently interrupting the physician. She is oriented to person, place, and time. She is able to recall three of three objects after 5 minutes and quickly and accurately performs 2-digit calculations. Which of the following is the most likely diagnosis?

- ☐ A) Alcohol withdrawal
- ☐ B) Bipolar disorder
- ☐ C) Cocaine abuse
- ☐ D) Delusional disorder, persecutory type
- ☐ E) Inhalant abuse
- ☐ F) Paranoid personality disorder
- ☐ G) PCP abuse
- ☐ H) Schizophrenia, paranoid type

40. A 15-year-old girl is brought to the emergency department 90 minutes after falling during ballet practice. She says that she has recently felt weak but has not had fever or other symptoms. She reports that she has felt "down" and "disgusted" with how she looks for the past 5 months. During this time, she has had difficulty sleeping and a decreased appetite resulting in a 6.8-kg (15-lb) weight loss. Menarche has not yet occurred. She is not sexually active and does not drink alcohol or use illicit drugs. She appears younger than her stated age. She is 163 cm (5 ft 4 in) tall and weighs 42 kg (92 lb); BMI is 16 kg/m². Physical examination shows tenderness to palpation over the right tibia. On mental status examination, she makes poor eye contact, speaks softly, and answers questions with vague responses. Her leukocyte count is 2800/mm³, and serum triiodothyronine (T₃) concentration is 95 ng/dL. An x-ray of the right lower extremity shows a tibial stress fracture and significant osteoporosis. Which of the following is the most likely diagnosis?

- ☐ A) Anorexia nervosa
- ☐ B) Body dysmorphic disorder
- ☐ C) Hypoparathyroidism
- ☐ D) Hypothyroidism
- ☐ E) Major depressive disorder
- ☐ F) Occult malignancy

41. A 67-year-old woman, whose husband died 18 months ago, is admitted to the hospital because of severe headaches for 4 weeks. She is quiet and seems sad. She believes that she has brain cancer, deserves to be punished for her sins, and is being punished by God for past misdeeds. She has had fatigue and a 6.8-kg (15-lb) weight loss over the past 4 months. Physical and neurologic examinations show no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Adjustment disorder with mixed disturbance of emotions and conduct
- ☐ B) Delusional disorder
- ☐ C) Major depressive disorder
- ☐ D) Schizophrenia
- ☐ E) Somatization disorder

42. A 15-year-old boy is brought to the emergency department by his two friends because of unusual behavior that began at a party 1 hour ago. His friends say that he became raucous and made an unwanted sexual advance to a girl he had just met. He then ran into the street, careless to oncoming cars, saying he was able to avoid them. His pulse is 84/min, respirations are 12/min, and blood pressure is 110/70 mm Hg. He is agitated when left alone, but he calms down when spoken to by nursing staff. Physical examination shows an unsteady gait. Pupils are equal and reactive. Which of the following is the most likely drug taken?

- ☐ A) Alcohol
- ☐ B) Amphetamine
- ☐ C) Cocaine
- ☐ D) Heroin
- ☐ E) LSD
- ☐ F) Marijuana

43. A 26-year-old man comes to the physician because of increasing daytime sleepiness over the past 3 years. He has begun to unexpectedly fall asleep at work and is concerned that he will lose his job if he does not change his behavior. He has stopped driving because of fear that he will fall asleep at the wheel. He generally sleeps well at night. He states that he avoids exercise and caffeine ingestion in the evening and does not read or watch television in bed. His wife states that he does not snore or have restless leg movements while sleeping. He has vivid dream-like images just before falling asleep and on awakening. He describes his mood as generally good despite his worry about his job security. Polysomnography is most likely to show which of the following?

- ☐ A) Decreased REM density
- ☐ B) Decreased REM latency
- ☐ C) Decreased stage 1 sleep
- ☐ D) Increased sleep efficiency
- ☐ E) Increased sleep latency

44. A 52-year-old woman comes to the physician for a routine follow-up examination. She has received inpatient psychiatric treatment several times since the age of 25 years for auditory hallucinations and the belief that her thoughts and movements were being controlled by a local television station; her last admission to the hospital was 10 years ago. Treatment with haloperidol for the past 20 years has decreased the occurrence of symptoms. She attempted suicide at the ages of 30 and 38 years. She lives in a supervised residence and does not work. During the examination, she repeatedly smacks her lips and slightly protrudes her tongue. When the physician asks if she is chewing gum, she laughs and opens her mouth to show she is not. She can hold her mouth and tongue still when asked but begins lip smacking when the physician resumes the examination. Which of the following is the most appropriate next step in pharmacotherapy?

- ☐ A) Continue haloperidol and add alprazolam
- ☐ B) Continue haloperidol and add propranolol
- ☐ C) Discontinue haloperidol
- ☐ D) Discontinue haloperidol and begin chlorpromazine
- ☐ E) Discontinue haloperidol and begin risperidone

45. A 27-year-old woman comes to the physician for an examination prior to starting a new job. She has a 10-year history of binge-eating and self-induced vomiting. She takes no medications. She does not drink alcohol or use illicit drugs. She is 178 cm (5 ft 10 in) tall and weighs 72 kg (160 lb); BMI is 23 kg/m². Her temperature is 37°C (98.6°F), pulse is 70/min, respirations are 10/min, and blood pressure is 120/70 mm Hg. Physical examination shows yellow dental enamel and abdominal striae. Serum studies are most likely to show which of the following abnormalities?

- ☐ A) Decreased bicarbonate concentration
- ☐ B) Increased amylase activity
- ☐ C) Increased magnesium concentration
- ☐ D) Increased potassium concentration
- ☐ E) Increased sodium concentration

46. A 25-year-old man comes to the physician with his wife because of bizarre behavior for the past 3 months. She reports that he will only travel to work by taking certain routes and is overly concerned about air quality, routinely inspecting for dust on the furniture and closing the windows when he returns from work. He has organized his office and his home in a particular way and becomes extremely upset when anything is rearranged. He realizes that his demands are unreasonable but says that "I would be very nervous and tense if it were arranged any other way." He has had no changes in sleep pattern. Physical examination shows no abnormalities. On mental status examination, he reports feeling mildly anxious and depressed "all the time." Results of laboratory studies are within reference ranges. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Bupropion
- ☐ B) Buspirone
- ☐ C) Clonazepam
- ☐ D) Haloperidol
- ☐ E) Sertraline

47. A 72-year-old woman with metastatic breast cancer is admitted to the hospital because of changes in her mental status during the past 3 days. Her family says she has been mumbling and softly moaning. They also report that she has been drinking more alcohol since she found out how far her disease has progressed. Current medications include fentanyl patches. There have not been any recent changes to her medication regimen. Vital signs are within normal limits. Physical examination shows no abnormalities. On mental status examination, she appears to be hallucinating. She is not oriented to person, place, or time. Laboratory studies show:

Hematocrit	35%
Leukocyte count	8000/mm ³
Serum	
Na ⁺	137 mEq/L
Ca ²⁺	12.5 mg/dL
Urea nitrogen	15 mg/dL
Creatinine	1.1 mg/dL
Albumin	1.5 g/dL

Urinalysis shows no abnormalities. X-rays of the chest and a CT scan of the head show no abnormalities. Which of the following is the most likely cause of this patient's change in mental status?

- ☐ A) Adverse effect of fentanyl
- ☐ B) Alcohol withdrawal
- ☐ C) Hypercalcemia
- ☐ D) Major depressive disorder with psychotic features
- ☐ E) Meningitis

48. A 37-year-old man is brought to the emergency department by the police after they found him sitting in an alley unaware of his surroundings or situation. On arrival, he cannot remember his name, date of birth, or age. Vital signs are within normal limits. Examination shows numerous ecchymoses over the face and extremities. Right lateral gaze is severely restricted. On left lateral gaze, there is bilateral nystagmus. There is mild gynecomastia and palmar erythema. Abdominal examination shows right upper quadrant tenderness. The liver edge is palpated 4 cm below the right costal margin. Muscle strength is normal, but he is unable to stand or walk. Deep tendon reflexes are absent. The most likely cause of these findings is a deficiency of which of the following?

- ☐ A) Folic acid
- ☐ B) Niacin
- ☐ C) Vitamin B₁ (thiamine)
- ☐ D) Vitamin B₆
- ☐ E) Vitamin B₁₂ (cobalamin)

49. A 16-year-old girl is brought to the physician by her father because of unusual behavior since returning from a party early this morning. Her father reports that when she arrived home, she barricaded herself in her room and refused to respond to his pleas to open the door. When he entered the room 1 hour later, he found her huddled in a corner in the dark, saying that the police wanted to kill her. She has no history of medical or psychiatric illness. Her pulse is 115/min, and blood pressure is 150/90 mm Hg. The pupils are 2 mm, equal, and reactive to light and accommodation. The conjunctivae are injected. On mental status examination, she appears apprehensive but is cooperative. Her speech is slow with a prolonged response latency. She believes that she is being followed by plainclothes police officers who want to assassinate her. She hears a male voice making threatening comments. She is oriented to person and place but not to time. She can perform double-digit money calculations but not serial sevens. Which of the following is the most likely cause of this patient's symptoms?

- ☐ A) Alcohol intoxication
- ☐ B) Alcohol withdrawal
- ☐ C) Cocaine withdrawal
- ☐ D) Delusional disorder, persecutory type
- ☐ E) Marijuana intoxication
- ☐ F) Marijuana withdrawal
- ☐ G) Panic disorder
- ☐ H) Paranoid personality disorder
- ☐ I) Schizophrenia, paranoid type



50. A 29-year-old man comes to the physician because of the inability to maintain an erection through completion of sexual activity. He has been married for 3 months and has not been able to complete coitus over the past 2 months. He is usually able to attain an erection by direct penile stimulation and is often aware of nocturnal erections when waking during the night. For the past month, he and his wife have attempted to have intercourse about twice a week. He drinks several alcoholic beverages before engaging in sexual activity in an effort to relax. He has had two major depressive episodes over the past 4 years and was successfully treated with desipramine; he has not taken the medication since his marriage. He has borderline hypertension but takes no medications for it. He has a strong family history of type 2 diabetes mellitus. He takes ranitidine for stomach problems. His vital signs are normal. Cranial nerve function appears normal. He walks normally. Laboratory studies show:

Hemoglobin	11 g/dL
Platelet count	178,000/mm ³
Serum	
Fasting glucose	86 mg/dL
γ-Glutamyltransferase	70 U/L (N=5–50)

Which of the following is the most likely explanation for these findings?

- ☐ A) Alcohol use
- ☐ B) Desipramine therapy
- ☐ C) Diabetes mellitus
- ☐ D) Generalized anxiety disorder
- ☐ E) Major depressive disorder
- ☐ F) Marital conflict
- ☐ G) Ranitidine therapy